COMP1870 Theoretical Foundations of Computer Science I

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Solutions to worksheet for week 1.2

1. In this question, assume that the domain is the set of positive integers and use the following predicates:

$$\begin{aligned} P(x) &\text{ means } x \text{ is prime} \\ Q(x) &\text{ means } x \text{ is a power of a prime number} \\ S(x) &\text{ means } x \text{ is a perfect square} \\ D(x,y) &\text{ means } x \text{ divides } y \\ x = y &\text{ means } x \text{ is equal to } y \end{aligned}$$

   (a) Translate the following statement into predicate logic:

   > There exists a positive integer x such that x is a perfect square and there exist positive integers y and z such that y is a prime and z is a prime and y is not equal to z and y divides x and z divides x.

   Is this statement true or false?

   Solution. $\exists x\,(S(x) \wedge \exists y\,\exists z\,(P(y) \wedge P(z) \wedge \neg(y = z) \wedge D(y,x) \wedge D(z,x)))$

   In order to make this true, we need to find a perfect square x. We need two prime numbers y and z such that y is not equal to z. We want y and z to both divide x. So in summary, we are looking for a perfect square, and two distinct prime numbers that divide it. So let $x = 36 = 6^2$ and let $y = 2$ and let $z = 3$. This shows that there does exist such a choice of x, y, and z. Therefore the statement is true.

   (b) Translate the following statement into predicate logic:

   > For every positive integer x, if x is a prime power then there exists a positive integer y such that y is prime and for every positive integer z, if z divides x and z is prime then z is equal to y.

   Is this statement true or false?

   Solution. $\forall x\,(Q(x) \Rightarrow (\exists y\,P(y) \wedge \forall z\,(D(z,x) \wedge P(z) \Rightarrow z = y)))$

   This is saying that whenever x is a prime power, then there is some prime y such that if z is prime and z divides x, then z equals y. So imagine that x is a prime power. Then $x = p^t$ where p is a prime and t is a positive integer. Now we can set y to be p. Whenever z is a prime that divides x, z must be equal to $p = y$, since there is only one prime that divides x. Therefore z divides y. So this statement is true.

   (c) Translate the following statement into English:

$$\forall x\,(Q(x) \Rightarrow \forall y\,\forall z\,(P(z) \wedge P(y) \wedge D(y,x) \wedge D(z,x) \Rightarrow y = z)$$

   Is this statement true or false?

   Solution. For every positive integer x, if x is a prime power, then for every pair of positive integers y and z, if y and z are primes that both divide x, then y is equal to z.

   Since a prime power is divided by only one prime, assuming that x is a power of the prime p, and y and z are primes that divide x, then $y = z = p$. So this statement is true.

   (d) Translate the following statement into English:

$$\exists x\,(S(x) \wedge P(x))$$

Is this statement true or false?

Solution. There exists a positive integer x such that x is a perfect square and a prime.

This can't be true, because a prime cannot be written as an integer times itself (by the definition of a prime).

2. Use the logical equivalences in Sections 2.1 and 2.2 of the lecture notes to show that

$$\neg\forall x\,(P(x) \Rightarrow \exists y\,(P(y) \vee \neg Q(x))) \qquad \text{is equivalent to} \qquad \exists x\,(P(x) \wedge \forall y\,(\neg P(y) \wedge Q(x))).$$

Solution.

$$\begin{array}{lcll}
\neg\forall x\,(P(x) \Rightarrow \exists y\,(P(y) \vee \neg Q(x))) & \Leftrightarrow & \exists x\,\neg(P(x) \Rightarrow \exists y\,(P(y) \vee \neg Q(x))) & \text{(Proposition 2.8)} \\
& \Leftrightarrow & \exists x\,(P(x) \wedge \neg(\exists y\,(P(y) \vee \neg Q(x)))) & \text{(Proposition 2.7)} \\
& \Leftrightarrow & \exists x\,(P(x) \wedge \forall y\,\neg(P(y) \vee \neg Q(x))) & \text{(Proposition 2.9)} \\
& \Leftrightarrow & \exists x\,(P(x) \wedge \forall y\,(\neg P(y) \wedge \neg\neg Q(x))) & \text{(De Morgan's law)} \\
& \Leftrightarrow & \exists x\,(P(x) \wedge \forall y\,(\neg P(y) \wedge Q(x))) & \text{(Double negation)}
\end{array}$$

4. Use equivalences to find a statement in predicate logic which is equivalent to

$$\neg(\forall x\,((\exists y\,Q(y)) \Rightarrow (\exists z\,R(z))))$$

and where every negation sign appears directly before a predicate.

Solution.

$$\begin{array}{lcll}
\neg(\forall x\,((\exists y\,Q(y)) \Rightarrow (\exists z\,R(z)))) & \Leftrightarrow & \exists x\,\neg((\exists y\,Q(y)) \Rightarrow (\exists z\,R(z))) & \text{(Proposition 2.8)} \\
& \Leftrightarrow & \exists x\,((\exists y\,Q(y)) \wedge \neg(\exists z\,R(z))) & \text{(Proposition 2.7)} \\
& \Leftrightarrow & \exists x\,((\exists y\,Q(y)) \wedge (\forall z\,\neg R(z))) & \text{(Proposition 2.9)}
\end{array}$$

5. By completing truth tables, decide whether the following arguments are valid.

(a) $P_1 \Rightarrow (P_2 \vee P_3)$, $\neg P_2 \;\therefore\; P_1 \Rightarrow P_3$

Solution.

P_1	P_2	P_3	$P_1 \Rightarrow (P_2 \vee P_3)$	$\neg P_2$	$P_1 \Rightarrow P_3$
T	**T**	**T**	**T**	**F**	**T**
T	**T**	**F**	**T**	**F**	**F**
T	**F**	**T**	**T**	**T**	**T**
T	**F**	**F**	**F**	**T**	**F**
F	**T**	**T**	**T**	**F**	**T**
F	**T**	**F**	**T**	**F**	**T**
F	**F**	**T**	**T**	**T**	**T**
F	**F**	**F**	**T**	**T**	**T**

There are three lines of the truth table where both of the premises are true. These are the third, seventh, and eighth lines. Because the conclusion is true in all of these lines, the truth of the premises forces the truth of the conclusion. This means that the argument is valid.

(b) $P_1 \Rightarrow (P_2 \vee P_3)$, $P_2 \therefore P_1 \Rightarrow P_3$

Solution.

P_1	P_2	P_3	$P_1 \Rightarrow (P_2 \vee P_3)$	P_2	$P_1 \Rightarrow P_3$
T	**T**	**T**	**T**	**T**	**T**
T	**T**	**F**	**T**	**T**	**F**
T	**F**	**T**	**T**	**F**	**T**
T	**F**	**F**	**F**	**F**	**F**
F	**T**	**T**	**T**	**T**	**T**
F	**T**	**F**	**T**	**T**	**T**
F	**F**	**T**	**T**	**F**	**T**
F	**F**	**F**	**T**	**F**	**T**

There are four lines of the truth table where both of the premises are true. These are the first, second, fifth, and sixth lines. In at least one of these lines, the conclusion is not true (the second line). Therefore the truth of the premises *does not* force the truth of the conclusion. This means that the argument is invalid.

(c) $P_1 \Rightarrow P_2, P_1 \wedge (\neg P_2) \therefore P_1$

Solution.

P_1	P_2	$P_1 \Rightarrow P_2$	$P_1 \wedge (\neg P_2)$	P_1
T	**T**	**T**	**F**	**T**
T	**F**	**F**	**T**	**F**
F	**T**	**T**	**F**	**T**
F	**F**	**T**	**F**	**F**

There are no lines where the premises are all true. Therefore the argument is valid, by the principle of explosion.

6. The Fitch diagram below shows a proof that $P_1 \wedge (P_2 \vee P_3)$, $(P_2 \vee P_3) \Rightarrow P_4 \therefore P_1 \wedge P_4$ is a valid argument. Fill in all of the justifications on the right hand side.

1. $P_1 \wedge (P_2 \vee P_3)$	Premise
2. $(P_2 \vee P_3) \Rightarrow P_4$	Premise
3. P_1	$\wedge$Elim 1
4. $P_2 \vee P_3$	$\wedge$Elim 1
5. P_4	$\Rightarrow$ Elim $2, 4$
6. $P_1 \wedge P_4$	$\wedge$Intro $3, 5$

7. The Fitch diagram below shows a proof that $P_1 \Rightarrow (P_2 \wedge P_3)$, $(P_2 \wedge P_4) \Rightarrow P_5$, $P_1 \wedge P_4 \ \therefore \ P_5$ is a valid argument. Fill in all of the justifications on the right hand side.

1. $P_1 \Rightarrow (P_2 \wedge P_3)$	Premise
2. $(P_2 \wedge P_4) \Rightarrow P_5$	Premise
3. $P_1 \wedge P_4$	Premise
4. P_1	$\wedge$Elim 3
5. $P_2 \wedge P_3$	$\Rightarrow$ Elim $1, 4$
6. P_2	$\wedge$Elim 5
7. P_4	$\wedge$Elim 3
8. $P_2 \wedge P_4$	$\wedge$Intro $6, 7$
9. P_5	$\Rightarrow$ Elim $2, 8$

9. The Fitch diagram below shows a proof that $P_1 \Rightarrow P_2 \ \therefore \ (\neg P_1) \vee P_2$ is a valid argument. Fill in all of the justifications on the right hand side.

Solution.

1. $P_1 \Rightarrow P_2$	Premise
2. $\neg((\neg P_1) \vee P_2)$	Assumption
3. P_1	Assumption
4. P_2	$\Rightarrow$ Elim $1, 3$
5. $(\neg P_1) \vee P_2$	$\vee$Intro 4
6. $((\neg P_1) \vee P_2) \wedge \neg((\neg P_1) \vee P_2)$	$\wedge$Intro $5, 2$
7. $P_1 \Rightarrow ((\neg P_1) \vee P_2) \wedge \neg((\neg P_1) \vee P_2)$	$\Rightarrow$ Intro 3-6
8. $\neg P_1$	$\neg$Intro 7
9. $(\neg P_1) \vee P_2$	$\vee$Intro 8
10. $((\neg P_1) \vee P_2) \wedge \neg((\neg P_1) \vee P_2)$	$\wedge$Intro $9, 2$
11. $\neg((\neg P_1) \vee P_2) \Rightarrow ((\neg P_1) \vee P_2) \wedge \neg((\neg P_1) \vee P_2)$	$\Rightarrow$ Intro 2-10
12. $\neg\neg((\neg P_1) \vee P_2)$	$\neg$Intro 11
13. $(\neg P_1) \vee P_2$	$\neg\neg$ 12

10. Write a Fitch diagram which shows that

$$\neg(P_1 \wedge P_2) \ \therefore \ P_1 \Rightarrow (\neg P_2)$$

is a valid argument. (It is possible to do this in 8 lines.)
Solution.

1. $\neg(P_1 \wedge P_2)$	Premise
2. P_1	Assumption
3. P_2	Assumption
4. $P_1 \wedge P_2$	$\wedge$Intro $2, 3$
5. $(P_1 \wedge P_2) \wedge (\neg(P_1 \wedge P_2))$	$\wedge$Intro $4, 1$
6. $P_2 \Rightarrow ((P_1 \wedge P_2) \wedge (\neg(P_1 \wedge P_2)))$	$\Rightarrow$ Intro 3-5
7. $\neg P_2$	$\neg$Intro 6
8. $P_1 \Rightarrow (\neg P_2)$	$\Rightarrow$ Intro 2-7

11. Write a Fitch diagram which shows that

$$\neg(P_1 \vee P_2) \ \therefore \ (\neg P_1) \wedge (\neg P_2)$$

is a valid argument. (It is possible to do this in 12 lines.)
Solution.

1. $\neg(P_1 \vee P_2)$	Premise
2. P_1	Assumption
3. $P_1 \vee P_2$	$\vee$Intro 2
4. $(P_1 \vee P_2) \wedge (\neg(P_1 \vee P_2))$	$\wedge$Intro $3, 1$
5. $P_1 \Rightarrow ((P_1 \vee P_2) \wedge (\neg(P_1 \vee P_2)))$	$\Rightarrow$ Intro 2-4
6. $\neg P_1$	$\neg$Intro 5
7. P_2	Assumption
8. $P_1 \vee P_2$	$\vee$Intro 7
9. $(P_1 \vee P_2) \wedge (\neg(P_1 \vee P_2))$	$\wedge$Intro $8, 1$
10. $P_2 \Rightarrow ((P_1 \vee P_2) \wedge (\neg(P_1 \vee P_2)))$	$\Rightarrow$ Intro 7-9
11. $\neg P_2$	$\neg$Intro 10
12. $(\neg P_1) \wedge (\neg P_2)$	$\wedge$Intro $6, 11$

12. The following Fitch diagram shows a "proof" that the argument $P_1 \Rightarrow P_2 \ \therefore \ \neg P_1$ is a valid argument.

1. $P_1 \Rightarrow P_2$	Premise
2. P_1	Assumption
3. P_2	$\Rightarrow$ Elim $1, 2$
4. $\neg P_2$	$\neg$Intro 3
5. $P_2 \wedge (\neg P_2)$	$\wedge$Intro $3, 4$
6. $P_1 \Rightarrow (P_2 \wedge (\neg P_2))$	$\Rightarrow$ Intro 2-5
7. $\neg P_1$	$\neg$Intro 6

(a) Verify that $P_1 \Rightarrow P_2 \ \therefore \ \neg P_1$ is not a valid argument by constructing truth tables.
Solution.

P_1	P_2	$P_1 \Rightarrow P_2$	$\neg P_1$
T	**T**	**T**	**F**
T	**F**	**F**	**F**
F	**T**	**T**	**T**
F	**F**	**T**	**T**

There are three lines where the premises are all true. (Actually, there is only one premise, so we just need to look for the lines where $P_1 \Rightarrow P_2$ is true.) There exists a line where the premise is true, but the conclusion is false. (This happens on line one.) So the truth of the premises does not force the truth of the conclusion. Therefore the argument is not valid.

(b) On what line of this Fitch diagram does an error occur? What is the error?
Solution. The error occurs on line 4. We can't use the negation introduction rule to introduce a negation whenever we feel like it. In order to justify introducing $\neg P_2$ we would need to show that P_2 implies a statement of the form $B \wedge (\neg B)$. We haven't done this, so we can't justify introducing $\neg P_2$.

13. (*) Use the rules of inference to prove that

$$(\neg P_1) \wedge (\neg P_2) \ \therefore \ \neg(P_1 \vee P_2)$$

is a valid argument. It is possible to do this in 18 lines. (Can you find a way to do it in fewer than 18?)

Solution.

1.	$(\neg P_1) \wedge (\neg P_2)$	Premise
2.	$\neg P_1$	$\wedge$Elim 1
3.	$\neg P_2$	$\wedge$Elim 1
4.	$P_1 \vee P_2$	Assumption
5.	P_1	Assumption
6.	$P_1 \wedge (\neg P_1)$	$\wedge$Intro $5, 2$
7.	$P_1 \Rightarrow (P_1 \wedge (\neg P_1))$	$\Rightarrow$ Intro 5-6
8.	P_2	Assumption
9.	$\neg P_1$	Assumption
10.	$P_2 \wedge (\neg P_2)$	$\wedge$Intro $8, 3$
11.	$(\neg P_1) \Rightarrow (P_2 \wedge (\neg P_2))$	$\Rightarrow$ Intro 9-10
12.	$\neg(\neg P_1)$	$\neg$Intro 11
13.	P_1	$\neg\neg$ 12
14.	$P_1 \wedge (\neg P_1)$	$\wedge$Intro $13, 2$
15.	$P_2 \Rightarrow (P_1 \wedge (\neg P_1))$	$\Rightarrow$ Intro 8-14
16.	$P_1 \wedge (\neg P_1)$	$\vee$Elim $4, 7, 15$
17.	$(P_1 \vee P_2) \Rightarrow (P_1 \wedge (\neg P_1))$	$\Rightarrow$ Intro 4-16
18.	$\neg(P_1 \vee P_2)$	$\neg$Intro 17

14. (*) Use the rules of inference to prove that

$$\neg(P_1 \wedge P_2) \ \therefore \ (\neg P_1) \vee (\neg P_2)$$

is a valid argument. It is possible to do this in 24 lines. (Can you find a way to do it in fewer than 24?)

Solution.

1. $\neg(P_1 \wedge P_2)$	Premise
2. $\neg(P_1 \vee \neg P_1)$	Assumption
3. P_1	Assumption
4. $P_1 \vee \neg P_1$	$\vee$Intro 3
5. $(P_1 \vee \neg P_1) \wedge (\neg(P_1 \vee \neg P_1))$	$\wedge$Intro 4, 2
6. $P_1 \Rightarrow ((P_1 \vee \neg P_1) \wedge (\neg(P_1 \vee \neg P_1)))$	$\Rightarrow$ Intro 3-5
7. $\neg P_1$	$\neg$Intro 6
8. $P_1 \vee \neg P_1$	$\vee$Intro 7
9. $(P_1 \vee \neg P_1) \wedge (\neg(P_1 \vee \neg P_1))$	$\wedge$Intro 8, 2
10. $\neg(P_1 \vee \neg P_1) \Rightarrow (P_1 \vee \neg P_1) \wedge (\neg(P_1 \vee \neg P_1))$	$\Rightarrow$ Intro 2-9
11. $\neg(\neg(P_1 \vee \neg P_1))$	$\neg$Intro 10
12. $P_1 \vee \neg P_1$	$\neg\neg$ 11
13. P_1	Assumption
14. P_2	Assumption
15. $P_1 \wedge P_2$	$\wedge$Intro 13, 14
16. $(P_1 \wedge P_2) \wedge (\neg(P_1 \wedge P_2))$	$\wedge$Intro 15, 1
17. $P_2 \Rightarrow ((P_1 \wedge P_2) \wedge (\neg(P_1 \wedge P_2)))$	$\Rightarrow$ Intro 14-16
18. $\neg P_2$	$\neg$Intro 17
19. $(\neg P_1) \vee (\neg P_2)$	$\vee$Intro 18
20. $P_1 \Rightarrow (\neg P_1) \vee (\neg P_2)$	$\Rightarrow$ Intro 13-19
21. $\neg P_1$	Assumption
22. $(\neg P_1) \vee (\neg P_2)$	$\vee$Intro 21
23. $(\neg P_1) \Rightarrow (\neg P_1) \vee (\neg P_2)$	$\Rightarrow$ Intro 21-22
24. $(\neg P_1) \vee (\neg P_2)$	$\vee$Elim 12, 20, 23